

# Simple summary

## Rural distribution and aggregation

Puapur, Purulia

### Bottom line

Promising, but verify local prices and demand and start small.

### Why this venture

It is the lowest-investment tested configuration that satisfies the active profile constraints and improves useful local flow. The evidence indicates a modelled service gap inside the selected catchment. This is a planning estimate, not measured locality turnover.

### Who this suits

An owner-operator who can supervise daily buying, selling, quality and cash control. Avoid it if you cannot verify local customers, suppliers, prices and licences before committing capital.

## Money needed

|                       |                              |                       |                         |
|-----------------------|------------------------------|-----------------------|-------------------------|
| Total project cost    | Own money used               | Money kept as reserve | External finance needed |
| INR 20,762 - 24,373   | INR 22,567 - 22,567          | INR 77,433 - 77,433   | INR 0 - 0               |
| Monthly revenue range | Monthly operating-cash range | Operating break-even  | Investment payback      |
| INR 1,610 - 1,968     | INR 626 - 938                | Month 1               | Month 32                |

## Local market

|                               |                                |                                                   |
|-------------------------------|--------------------------------|---------------------------------------------------|
| Demand opportunity            | Price guidance                 | Competition                                       |
| 19,095.0 - 35,462.2 INR/month | INR 0 - 1 / gross margin share | 6 direct / 0 indirect; LOW_PROXY_DENSITY; 20.0 km |

### Top advantages

- Uses the minimum-capital configuration selected by the MVV test.
- Keeps the recommendation tied to the selected locality and catchment.
- Shows cash resilience, break-even and failure conditions before investment.

### Main disadvantages

- Local demand and supply are modelled from available evidence, not audited shop sales.
- OSM candidate counts do not measure competitor capacity, sales or market share.

### Top risks

- Actual selling price or input cost may differ from the planning range.
- Working-capital pressure can delay break-even or create a cash shortfall.
- Customer and supplier availability must be checked on the ground.

### First actions

1. Validate at least 20 likely customers and three suppliers before paying for equipment.
2. Confirm current prices, licences, utilities and delivery costs in writing.
3. Start with the minimum viable setup, track weekly cash, and expand only after repeat demand.

Rural distribution and aggregation

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|                            |                                                      |
|----------------------------|------------------------------------------------------|
| Canonical locality         | Puapur, Purulia                                      |
| Locality type              | VILLAGE                                              |
| Project cost               | INR 22,567                                           |
| Own capital deployed       | INR 22,567                                           |
| Finance required           | INR 0                                                |
| confidence                 | MEDIUM                                               |
| Demand / opportunity       | 19,095.03 - 35,462.2 INR/month (MODELLED_BENCHMARK)  |
| Reachable incumbent supply | 11,738.75 - 21,800.54 INR/month (MODELLED_BENCHMARK) |
| Price / unit value         | 0.34 - 0.56 gross margin share (MODELLED_BENCHMARK)  |
| Planning radius            | 20 km                                                |

Confidence is MEDIUM. Historical, estimated and current evidence remain separately labelled in the technical report.

Customer segments

- small retailers
- producers
- institutions and small businesses

Supplier plan

- regional wholesalers
- producer groups
- transport contractors

|                                       |                             |
|---------------------------------------|-----------------------------|
| Direct OSM candidates inside radius   | 6                           |
| Indirect OSM candidates inside radius | 0                           |
| Competition intensity                 | LOW_PROXY_DENSITY           |
| Coordinates / quality                 | BLOCK_SIBLING_MEDIAN_PROXY  |
| Nearest market                        | Neamatpur Vegetable Market  |
| Nearest institution                   | Career Building Private ITI |

OSM candidates are deduplicated direct/indirect proxies; capacity, sales and market shares remain unknown, so HHI is not calculated.

All named direct alternatives

- Neamatpur Vegetable Market | MARKET | 9.07 km
- Galaxy mall | MARKET | 12.72 km
- Burnpur Daily Market | MARKET | 12.99 km
- Centrum Mall | MARKET | 13.99 km
- Adda Market | MARKET | 16.14 km
- Infinity Tower | MARKET | 17.01 km

Channels

- PRIMARY: fixed retailer route (LOW\_ASSUMPTION\_BASED)
- SECONDARY: aggregator service (LOW\_ASSUMPTION\_BASED)
- SECONDARY: institutional delivery (LOW\_ASSUMPTION\_BASED)

Operational factors

- fuel
- road access
- vehicle efficiency
- load factor
- empty return
- downtime

Weather and seasonality

- heavy rain/flood route disruption

|                     |            |
|---------------------|------------|
| Equipment and setup | INR 13,540 |
| Working capital     | INR 9,027  |
| Monthly fixed OPEX  | INR 23     |
| People              | 1          |
| Space               | 40 sq ft   |
| Service radius      | 9 km       |
| Inventory days      | 5.0        |
| Receivable days     | 10.0       |
| Payable days        | 7.0        |

Equipment

- rented/shared vehicle access
- handling crates
- route/phone tools

Quality controls

- trip sheet
- load-factor log
- fuel and maintenance log

Licences to verify

- commercial vehicle and trade compliance as applicable

CAPEX, OPEX and working capital

|                                   |           |
|-----------------------------------|-----------|
| Equipment And Fixtures            | INR 9,478 |
| Installation And Setup            | INR 2,031 |
| Premises Deposit And Basic Fitout | INR 1,354 |
| Licensing And Contingency         | INR 677   |

Insurance / protection

- commercial vehicle and goods-in-transit protection

|                      |           |
|----------------------|-----------|
| Monthly revenue      | INR 1,789 |
| Operating cash       | INR 782   |
| Operating break-even | Month 1   |
| Cash break-even      | Month 1   |
| Investment payback   | Month 32  |
| 36-month NPV         | INR -785  |
| Annualized IRR       | 9.5%      |
| Break-even volume    | 51.16     |

36-month closing cash

- Month 3: Revenue INR 894; Operating cash INR 379; closing cash INR 78,169
- Month 6: Revenue INR 1,789; Operating cash INR 782; closing cash INR 80,112
- Month 9: Revenue INR 1,789; Operating cash INR 782; closing cash INR 82,458
- Month 12: Revenue INR 1,789; Operating cash INR 782; closing cash INR 84,804
- Month 15: Revenue INR 1,789; Operating cash INR 782; closing cash INR 87,149
- Month 18: Revenue INR 1,789; Operating cash INR 782; closing cash INR 89,495
- Month 21: Revenue INR 1,789; Operating cash INR 782; closing cash INR 91,841
- Month 24: Revenue INR 1,789; Operating cash INR 782; closing cash INR 94,187
- Month 27: Revenue INR 1,789; Operating cash INR 782; closing cash INR 96,532
- Month 30: Revenue INR 1,789; Operating cash INR 782; closing cash INR 98,878
- Month 33: Revenue INR 1,789; Operating cash INR 782; closing cash INR 101,224
- Month 36: Revenue INR 1,789; Operating cash INR 782; closing cash INR 103,570

Possible finance fit

- Pradhan Mantri MUDRA Yojana: illustrative eligibility screening; not lender approval
- Animal Husbandry Infrastructure Development Fund: current rule-window eligibility screening through 2026-09-30; not lender approval or confirmation that the application window is accepting submissions

|                                 |             |
|---------------------------------|-------------|
| Scenarios                       | 512         |
| Remain solvent                  | 100.0%      |
| Payback within 36 months        | 58.2%       |
| 10th percentile minimum cash    | INR 77,498  |
| Worst 5% cumulative cash (CVaR) | INR -14,202 |

Scenario rates are modelled planning survival, not probability of success. The triangular factors are not empirically calibrated.

STRENGTHS

- The modelled cash-conversion cycle is short at 8 days.
- Month-12 operating cash flow is positive in the central model.

WEAKNESSES

- Local demand is a regional benchmark, not observed locality sales.

OPPORTUNITIES

- The planning graph contains 10509 units of unserved flow.
- A secondary aggregator service channel can diversify the primary route.
- 1 marginal capacity repair options were evaluated.

THREATS

- Selling Price is the strongest tested cash driver (elasticity 2.29).
- Competitor capacity and market shares remain unknown.

Adaptive failure boundaries

- monthly\_demand: - planning revenue units/month - No cash failure up to 100% demand deterioration.
- selling\_price\_factor: - share of central planning price - No cash failure up to 100% selling price deterioration.
- variable\_cost\_factor: 4.14 multiple of central variable cost - Plan becomes cash-negative above approximately 4.142x central variable cost.
- fixed\_opex\_factor: - multiple of central fixed OPEX - No cash failure up to 900% fixed OPEX increase.
- minimum\_cash\_buffer: 0 INR opening cash after startup investment - Approximately INR 0 opening cash is required to remain non-negative in the central 36-month model.

Sensitivity tornado

- selling\_price: LOW INR 692; MEDIUM INR 782; HIGH INR 871
- variable\_cost: LOW INR 831; MEDIUM INR 782; HIGH INR 733
- fixed\_opex: LOW INR 783; MEDIUM INR 782; HIGH INR 781
- demand: LOW INR 782; MEDIUM INR 782; HIGH INR 782

**Pre-mortem: why this could fail**

- Adverse selling price moved cash below plan. Validate current customer prices and negotiate supplier terms first.
- Adverse variable cost moved cash below plan. Obtain two supplier quotes and cap procurement cost per unit.
- Adverse fixed opex moved cash below plan. Avoid a long lease until three months of demand are demonstrated.
- Customer acquisition was slower because nearby alternatives existed. Interview customers and differentiate channel/service before launch.
- The regional benchmark did not translate into real locality sales. Run a small paid pilot and replace the benchmark with observed sales.

**Before starting**

- Validate at least two current supplier quotations and one customer selling price.
- Confirm premises, power/water/internet needs and every listed licence with authority.
- Ring-fence at least INR 9,027 as working capital.

**Week 1**

- Interview at least ten target customers and record purchase frequency and price.
- Run a paid micro-pilot before buying the full equipment configuration.

**Month 1**

- Track daily sales, contribution margin, inventory days, cash and rejected/wasted stock.
- Reconcile supplier invoices and customer receipts weekly.

**Months 2–3**

- Continue only if contribution margin and closing cash remain non-negative.
- Add the secondary channel only after the primary channel repeats reliably.

**Months 4–6**

- Compare realized sales with the model interval and recalibrate before expansion.
- Build at least two active suppliers and avoid single-buyer dependence.

**Stop / reconsider**

- Use the tested demand-deterioration statement as the sales stop rule.
- Validate current customer prices and negotiate supplier terms first.
- Obtain two supplier quotes and cap procurement cost per unit.

**Action plan**

- Stage 0: validate prices, suppliers and paid demand using shared/rented assets.
- Stage 1: deploy the selected configuration and preserve INR 2,257 reserve.
- Stage 2: the central model first supports a 70% utilization plus reserve trigger in month 5.
- Stop-rule context: No cash failure up to 100% demand deterioration.

|                |                                      |
|----------------|--------------------------------------|
| confidence     | MEDIUM                               |
| Status         | CONDITIONAL                          |
| Decision chain | decision-v6                          |
| Analysis ID    | 5ec1c679-b269-40d1-97a2-51bcbeaf8bae |

households\_observed\_2011 | Census 2011 Primary Census Abstract TV | 2011-03-01 | HISTORICAL\_BASELINE | HIGH

livestock\_population\_buffalo | 20th Livestock Census village and ward data | UNKNOWN | STALE\_FOR\_DECISION | HIGH

livestock\_population\_cattle | 20th Livestock Census village and ward data | UNKNOWN | STALE\_FOR\_DECISION | HIGH

livestock\_population\_goat | 20th Livestock Census village and ward data | UNKNOWN | STALE\_FOR\_DECISION | HIGH

livestock\_population\_pig | 20th Livestock Census village and ward data | UNKNOWN | STALE\_FOR\_DECISION | HIGH

livestock\_population\_sheep | 20th Livestock Census village and ward data | UNKNOWN | STALE\_FOR\_DECISION | HIGH

population\_female\_observed\_2011 | Census 2011 Primary Census Abstract TV | 2011-03-01 | HISTORICAL\_BASELINE | HIGH

population\_male\_observed\_2011 | Census 2011 Primary Census Abstract TV | 2011-03-01 | HISTORICAL\_BASELINE | HIGH

population\_observed\_2011 | Census 2011 Primary Census Abstract TV | 2011-03-01 | HISTORICAL\_BASELINE | HIGH

asuse\_nic10\_annual\_gva\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic10\_annual\_input\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic10\_annual\_output\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic10\_equipment\_investment\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic10\_total\_fixed\_assets\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic10\_workers\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_annual\_gva\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_annual\_input\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_annual\_output\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_equipment\_investment\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_total\_fixed\_assets\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic46\_workers\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_annual\_gva\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_annual\_input\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_annual\_output\_inr\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_equipment\_investment\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_total\_fixed\_assets\_owned\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

asuse\_nic47\_workers\_prior | DDI-IND-NSO-ASUSE-Jan2025-Dec2025 | 2025-12-31 | RECENT | MEDIUM

district\_annual\_milk\_production\_kg | West Bengal Estimated Milk Production 2024-25 | 2025-03-31 | RECENT | MEDIUM

monthly\_liquid\_milk\_expenditure\_inr\_per\_capita\_prior | DDI-IND-MOSPI-NSS-HCES23-24 | 2024-07-31 | RECENT | MEDIUM

monthly\_liquid\_milk\_expenditure\_inr\_per\_capita\_prior\_2022\_23 | DDI-IND-MOSPI-NSSO-HCES22-23 | 2023-07-31 | STALE\_FOR\_DECISION | MEDIUM

monthly\_liquid\_milk\_litres\_per\_capita\_prior | DDI-IND-MOSPI-NSS-HCES23-24 | 2024-07-31 | RECENT | MEDIUM

monthly\_liquid\_milk\_litres\_per\_capita\_prior\_2022\_23 | DDI-IND-MOSPI-NSSO-HCES22-23 | 2023-07-31 | STALE\_FOR\_DECISION | MEDIUM

#### Limitations

- Scheme category screening is current; actual rate, sanction and lender underwriting still require a lender quote.
- OSM POIs are volunteered proxy evidence and may be incomplete.

#### Sources

- <https://ard.wb.gov.in/upload/statistics/1772193244516.pdf>
- <https://censusindia.gov.in/nada/index.php/catalog/6531>
- <https://dahd.gov.in/sites/default/files/2023-07/VillageAndWardLevelDataMale-Female.xlsx>
- <https://financialservices.gov.in/pradhan-mantri-mudra-jojana-pmmy>
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- <https://microdata.gov.in/NADA/index.php/catalog/224>
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- <https://microdata.gov.in/NADA/index.php/catalog/293>
- <https://www.dahd.gov.in/sites/default/files/2026-07/TemporaryExtensionofSchemes.pdf>
- <https://www.openstreetmap.org/copyright>